Technical Product Definition & Machine Parts Final Assignment

Introduction to Mechanical Engineering
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Abstract

This assignment showcases the process of designing an output shaft for a gear transmission system used to power a cooling fan. The assignment includes designing the output shaft, coupling, pinion, keys, and bearings, as well as their mounting onto the shaft. Considerations for component positioning, and the inclusion of SolidWorks assembly and technical drawings are presented.

NOMENCLATURE

α	Pressure angle [deg]
α_a	Contact angle [deg]
α_c	Correction facotr
ϕ_d	face width factor
σ_F	bending stress [MPa]
σ_H	Contact stress [MPa]
σ_{-1}	Endurance limit in normal [MPa]
σ_{Flim}	Endurance limit for bending stress [MPa]
σ_{Hlim}	Endurance limit for contact stress [MPa]
σ_p	Allowable stress [MPa]
σ_s	Combination of shear and normal stress [MPa]
σ_s	Yield strength [MPa]
τ_{-1}	Endurance limit in shear [MPa]
a	center distance [mm]
A_L	Axial load [kN]
b	Face width [mm]
b_{key}	Width of key [mm]
C	Dynamic load capacity [kN]
C_0	Static load capacity [kN]
D	Shaft diameter [mm]
d	Gear diameter [mm]
f_p	friction coefficicent [kN]
f_t	friction coefficicent [kN]
F_a	Axial force [N]
F_r	Radial force [N]
F_t	Tangential force [N]
h_{key}	Height of key [mm]
$i = u$	Velocity ratio
I	Area moment of inertia [kgm2]
j	Number of gear theeth meshing per revolution
K	Load factor
K_{NF}	life factor for bending stress
K_{NH}	life factor for contact stress
K_t	Stress concentration factor
L_h	Lifetime [h]
l_{key}	Length of key [mm]
M	Bending Moment [Nmm]
m	module [mm]
N	Number of load cycles
n	Rotational speed [rpm]
P	Equivalent dynamic load [kN]
P_{motor}	Electric motor power [W]
R	Reaction force [N]
r	Radius [mm]
R_L	Radial load [kN]
S_F	Saftey factor for bending strength
S_H	Saftey factor for contact strength
T_i	Input Torque [Nmm]
V	Shear stress [N]
v	Pitch line velocity [m/s]
W	Section modulus [kgm2]
X	Radial load coefficicent [kN]
Y	Axial load coefficicent [kN]
Y_{Fa}	Tooth form factor
Y_{Sa}	Stress correction factor
Y_{ST}	Stressting correction factor for testing
z	The number of teeth in Spur gear
Z_E	Elastic Ceofficicent [MPa$^{1/2}$]
Z_H	Zone factor

I. Introduction

Analysis and design are indispensable aspects in the process of machine design. This paper aims to document the analysis and design process on a gear transmission used to power a cooling fan at a shipyard. The design of the gear transmission's output shaft including coupling, pinion, keys and bearings are explored in greater depth. More details can be found in the assignment file [1].

The design process showcased here stems primarily from two sources; Firstly, the lecture slides provided in the course *Introduction to Mechanical Engineering* [2], and the book *Analysis and Design of Machine Elements* by Wei Jiang [3].

Furthermore, this document has been structured like the recommended design process in [3], and consists of four parts. Design task identification, Assumptions and Decisions, Design analysis and Evaluation and lastly, Drawings and documentation. Although the report is structured in this format, the process is iterative, meaning multiple calculation cycles have been completed before reaching the final dimensions.

II. Design Task Identification

With the task at hand, there are a few specifications that can be quantified and implemented during the design phase to ensure the design is suitable for its application. The primary goal of the gear transmission is to increase the rotational speed of the output shaft in comparison to the input shaft. This requirement is specified by the given Train Value of two. Furthermore, the smaller gear known as the pinion is limited to having between 20-30 teeth. With this information, the speed ratio i can be defined as seen in equation 1, where the subscript p refers to the pinion and g to the gear.

$$i = u = \frac{n_p}{n_g} = \frac{d_g}{d_p} = \frac{z_g}{z_p} = \frac{T_g}{T_p} \tag{1}$$

The aim of the gear transmission is to increase the rotational speed, so the pinion is placed onto the output shaft. With this set-up shown in Fig.1, the speed ratio is given by the assignment and hence, $u = 2$. The transmission gearbox requirements are shown in table I

Gear ratio - 2:1		
	Input	Output
Speed [rpm]	1000	2000
Torque [Nm]	382	191

TABLE I: Transmission gearbox requirements

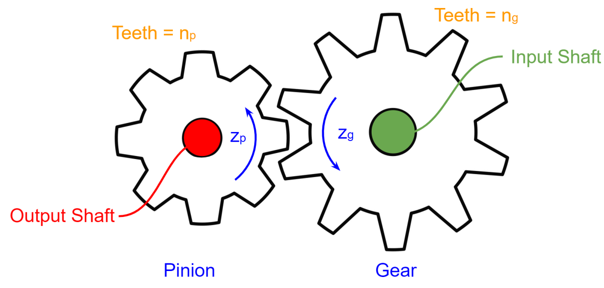

Fig. 1: Gear-Set up and definition

Another aspect that can be identified is the intended lifetime use of the device. With the operation time being 24hrs/day for 200 days per year, and minimal lifetime being 5 years, the lifetime L_h, can be determined as follows;

$$\begin{aligned} L_h &= 5 \cdot 200 \cdot 24 \\ &= 24000 \text{ [h]} \end{aligned}$$

Lastly, the last identifiable design specification is the input torque T_i, which can be determined from the 40 kW electrical motor with 1000 revs/min, as follows;

$$\begin{aligned} T_i &= \frac{60 P_{motor}}{2\pi n_g} \\ &= 382 \text{ [Nm]} \end{aligned}$$

It is important to note that the coupling is elastic because it allows for the assumption that the all the power from the electric motor is converted into torque. No loss of energy.

III. Assumptions and Decisions

Numerous decisions must be made when designing such a system as a gear transmission box. Among these, selecting an adequate material and surface treatment can be considered as one of the most important. The material must be able to withstand the applied loads, making it a vital part of the design process. Choosing the material was made with iterations, however it will be presented as pre-made choice in the report. All the important material properties are listed when used.

A. Material choice

When designing the spur gears, the choice was made to go with steel alloy Cr40 (AISI 5140) for the pinion [4]. The reason for this is because of the greater hardness of 280 HV. Calculations of the spur gear with a more standard steel, such as carbon steel (AISI 1045) yielded unsatisfactory parameters. For the other gear, the standard carbon steel 45 (AISI 1045) with hardness 210 HV was sufficient. When considering the shaft, the standard Carbon steel 45 (AISI 1045) was once again chosen, no need for a harder material was identified.

B. Surface treatment

All the materials considered are recommended to be heat treated by quenching and tempering to increase the hardness, strength and ductility [5]. According to [4], the hardness of carbon steel 45 (AISI 1045) can be increased to 245 HV by heat treatment, which is recommended.

C. Assumptions

During this design process, ideal working conditions are assumed. This means that there are negligible thermal effects, constant material properties and uniform loading. All the elements are also considered to be ideal, meaning no defects or varied behaviour.

IV. Design Analysis and Evaluation

The process begins by analysing gears. All the necessary gear dimensions can be determined from the preliminary information, and gear parameters must be used in further calculations of the shaft. The coupling can be selected after the gear, because its dimensions are needed for the shaft calculations, and it is independent of the shaft. The keys and bearings however, must are calculated simultaneously with the shaft, and are presented in the report as such.

A. Gears

The goal of this procedure is to design a safe and long-lasting gear drive. This process involves a few approximations and iterations. Namely, the standard pressure angle α of 20°for spur gears is assumed and a standard value for the modulus m is sought after. Furthermore, ideal working conditions are assumed, so no sliding friction, a constant velocity ratio and most importantly, perfect involute teeth profiles that ensure a uniform stress distribution.

The design procedure for this spur gear is split into two parts, one takes into account the gear teeth surface strength and the other the gear teeth bending strength, as suggested by [3]. For both of these sections, an initial number of teeth for the two gears must be selected. In general, gears with more teeth run smoother and quieter than less. Gears with fewer teeth may also endure interference or undercutting[1], and at high operating hours, as the case here, they may also introduce pitting[2]. For this reason, a higher number of teeth is preferred. A designer instinct recommend by [2] is to select a prime number of teeth for the pinion to avoid a hunting tooth that would impact the efficiency of the transmission.

For the specified range of 20-30 teeth for the pinion, the prime 29 has been selected over the alternative 23 for the reasons listed above. Using the speed ratio $u = 2$ and equation 1, the number of teeth for the gear is found to be $z_g = z_p u = 58$.

1) Design by the tooth surface strength of the pinion: Excessive contact stress is the reason for surface fatigue pitting [3]. To prevent pitting, the pitch diameter should be chosen so that it does not exceed the allowable stress $|\sigma_H|$ of the material, as described by equation 3, that is derived from the *Hertz Formula* [3].

$$|\sigma_H| = Z_H Z_E \sqrt{\frac{KF_t}{bd_p} \cdot \frac{u \pm 1}{u}} \tag{2}$$

$$d_p = \sqrt[3]{\frac{2KT_1}{\phi_d} \cdot \frac{u \pm 1}{u} \cdot \left(\frac{Z_H Z_E}{|\sigma_H|}\right)^2} \tag{3}$$

Force transmitted, F_t needed to solve for the stresses, is given by the following equation 4.

$$F_t = \frac{2T_i}{d} \tag{4}$$

The allowable contact stress is set by the endurance limit of the pinion and gear materials. The endurance limits are properties obtained from fatigue tests and depend on the accuracy grades of the spur gears. These values are given in Table 8.6 from [3], and make the following assumptions to ensure a 99% reliability; module $m = 3 \sim 5$ [mm], centre distance $a = 100$ [mm], face width $b = 10 \sim 50$[mm] and linear velocity $v = 10[\mathrm{ms}^{-1}]$. Based on the materials chosen in section Assumptions and Decisions, the contact endurance limits are extrapolated and can be seen in table II.

Material	Type	Hardness	σ_{Hlim} [MPa]
Hardened wrought steels	Alloy Steel	280 HV	740
	Carbon Steel	245 HV	575

TABLE II: Contact endurance limits for chosen materials from [3]

When the design criteria are different from the assumptions made to obtain the values in table II, a life factor K_{HN} and safety factor S_H are introduced, as shown in equation 5.

$$|\sigma_H| = \frac{K_{HN}\sigma_{Hlim}}{S_H} \tag{5}$$

Life factor K_{HN} is determined by calculating the number of load cycles N, given by the following equation 6, where the number of gear teeth meshing per revolution j is set to one.

$$N = 60njL_h \tag{6}$$

$$N_p = 60 \cdot 2000 \cdot 1 \cdot 24000 = 2.4 \times 10^7$$
$$N_g = 60 \cdot 1000 \cdot 1 \cdot 24000 = 1.2 \times 10^7$$

When these values are lower than, 5×10^7 as is the case here, a K_{HN} factor larger than one is selected based on the heat treatment applied. For this reason $K_{HNp} = K_{HNg} = 1.20$

[1] A condition generated when the curve of the gear tooth lies inside a line drawn tangent to the working profile

[2] A Common failure mode as a result of fatigue

have been chosen. The usual selection of $S_H = 1.0$ has also been selected, yielding the following:

$$|\sigma_{Hp}| = \frac{1.20 \cdot 740}{1} = 888 \text{ [MPa]}$$
$$|\sigma_{Hg}| = \frac{1.20 \cdot 575}{1} = 690 \text{ [MPa]}$$

Another value that needs to be determined in order to solve for the pitch diameter is the face width factor $\phi_d = \frac{b}{d}$. It is recommended by [6] to select a face width factor $\phi_d \leq 1.4$ for symmetrical support, such as the one in this application. It is also beneficial to select smaller gear tooth widths b for hardened surfaces [2], therefore a value of 1.1 has been chosen.

The elastic coefficient Z_E and zone factor Z_H for steel spur gears are 189.8 [MPa]$^{1/2}$ and 2.5 respectively [3]. It is important to note that the elastic coefficient Z_E depends on Young's E modulus and the Poisson's ratio, which do not change even if the metal is alloyed. Therefore, both gears will have the same coefficient despite their different steel composition [5]. Equation 3 is now solved with a trial factor K of 1.3, for the lower contact strength $|\sigma_{H2}|$.

$$d_p = \sqrt[3]{\frac{2 \cdot 1.3 \cdot 382000}{1.1} \cdot \frac{2+1}{2} \cdot \left(\frac{2.5 \cdot 189.8}{690}\right)^2}$$
$$= 86.20 \text{ [mm]}$$

The pitch line velocity v can also be solved as follow:

$$v = \frac{\pi d_p n_p}{60 \cdot 1000}$$
$$= \frac{\pi \cdot 86.20 \cdot 1000}{60 \cdot 1000}$$
$$= 4.51 \text{[m/s]}$$

This calculation is sufficient in determining a diameter that will withstand the applied load, however it is not the most optimal due to the assumed value of the load factor. The theory behind adjusting this load factor is outside the scope of this report, but the calculations have been conducted and can be found in the Appendix. The load factor was calculated to be $K = 1.532$, resulting in the adjusted diameter $d_p = 87.75$ [mm]. It will be considered instead of the $d_p = 86.20$ [mm] diameter going forward.

Lastly, the module of the gear can be computed using the diameter and number of gears as follows:

$$m = \frac{d_p}{z_p} = \frac{87.75}{29}$$
$$= 3.03 \text{ [mm]} \approx 3 \text{ [mm]}$$

2) Design by the tooth bending strength of the pinion: Gear's teeth can break due to the bending fatigue phenomenon caused by the bending action of a load [3]. To prevent tooth breakage, the bending stress should not exceed the allowed value of the material. Fatigue crack and failure tend to occur at the tip of the teeth and is described by the Lewis formula. It can be rewritten to solve for the module m shown in equation 8.

$$|\sigma_F| = \frac{KF_t}{bm} Y_{FA} Y_{SA} \tag{7}$$

$$m = \sqrt[3]{\frac{2KT_1}{\phi_d z_p^2} \cdot \left(\frac{Y_{Fa} Y_{Sa}}{|\sigma_F|}\right)^2} \tag{8}$$

The same as for contact stress, the endurance limit for the chosen materials can be found in table 8.6 from [3]. Consider the bending endurance limits shown in table III.

Material	Type	Hardness	σ_{Flim} [MPa]
Hardened wrought steels	Alloy Steel	280 HV	306
	Carbon Steel	245 HV	222

TABLE III: Bending endurance limits for chosen materials from [3]

For the same reasoning as for contact stress, the endurance limits are adjusted by life factors K_{FNp} and K_{FNg}, shown in equation 9.

$$|\sigma_F| = \frac{K_{FN} \sigma_{Flim}}{S_F} Y_{ST} \tag{9}$$

This time since, the load cycles are greater than 3×10^6, factors $K_{FN1} = K_{FN2} = 0.85$. Furthermore, for spur gears, the stress correction factor is set to two, and safety factor is set to 1.4 as recommended by [3]. Consider:

$$|\sigma_{Fp}| = \frac{0.85 \cdot 306}{1.4} 2 = 372 \text{ [MPa]}$$
$$|\sigma_{Fg}| = \frac{0.85 \cdot 222}{1.4} 2 = 270 \text{ [MPa]}$$

To consider the effect of other factors such as stress concentration at the root of the tooth fillet, or the root's bending stress, the stress correction factor Y_{SA}, is introduced. Furthermore, tooth form factor Y_{FA} considers the effect of the tooth's form and tip loading [3]. Both of these values can the extrapolated from table 8.4 shown in [3] for a profile shift coefficient $x = 0$. The profile shift is the amount by which the gear teeth are shifted relative to the standard involute profile. The extrapolated values are shown in table IV.

	Y_{Fa}	Y_{Sa}
Pinion	2.692	1.574
Gear	2.364	1.688

TABLE IV: Tooth form factor and stress correction factor interpolated from table 8.4 [3]

Now the module can be computed using equation 8, for the lower stress, which is for the gear. Consider:

$$m = \sqrt[3]{\frac{2 \cdot 1.3 \cdot 382000}{1.1 \cdot 29^2} \cdot \left(\frac{2.364 \cdot 1.688}{222}\right)^2}$$
$$= 2.68 \text{ [mm]} \approx 3 \text{ [mm]}$$

3) Geometrical Calculations: From the two modules obtained by the different analyses, it can be seen that the contact strength yields a larger module than the bending strength. The module is more closely related to the tooth bending stress, while the pitch diameter is relevant to contact strength [3]. Therefore, the module from bending strength is taken and rounded 2.68 [mm] ≈ 3, while the pitch diameter of 87.75 is taken and rounded to 87 mm. This gives the following number of teeth for the gears.

$$z_p = \frac{d_p}{m} = \frac{87}{3}$$
$$= 29$$
$$z_g = uz_p = 2 \cdot 29$$
$$= 58$$

Consider the final gear dimensions shown in table V.

	m = 3.0		
	z	d[mm]	b[mm]
Pinion	29	87	95
Gear	58	174	90

TABLE V: Final Gear dimensions

Lastly, the centre distance a between the two gears is given:

$$a = (d_p + d_g)/2 = (87 + 174)/2$$
$$= 130.5 \text{ [mm]}$$

This concludes the gear calculations.

4) Verification: In order to verify the design, the final gear dimensions are used in equations 2 and 7 to check whether the exerted stress is below the material's yield strength.
Contact stress:

$$|\sigma_{Hp}| = 2.5 \cdot 189.8 \sqrt{\frac{1.532 \left(\frac{2 \cdot 38200}{87}\right)}{95 \cdot 87} \cdot \frac{2+1}{2}}$$
$$= 728 \text{ [MPa]} < 888 \text{ [MPa]}$$
$$|\sigma_{Hg}| = 2.5 \cdot 189.8 \sqrt{\frac{1.532 \left(\frac{2 \cdot 191000}{174}\right)}{90 \cdot 174} \cdot \frac{2+1}{2}}$$
$$= 535 \text{ [MPa]} < 690 \text{ [MPa]}$$

Contact stress criteria are met.
Bending stress:

$$|\sigma_{Fp}| = \frac{1.499 \cdot \left(\frac{2 \cdot 38200}{87}\right)}{95 \cdot 3} 2.692 \cdot 1.574$$
$$= 334 \text{ [MPa]} < 372 \text{ [MPa]}$$
$$|\sigma_{Fg}| = \frac{1.499 \cdot \left(\frac{2 \cdot 191000}{174}\right)}{95 \cdot 3} 2.364 \cdot 1.688$$
$$= 258 \text{ [MPa]} < 270 \text{ [MPa]}$$

Bending stress criteria are met. With these results, the gear calculations are verified.

B. *Coupling*

A coupling is a device that connects two shaft ends to transmit power and torque. In gear transmission at hand, two couplings are required [3]. One from the electric motor to the input shaft and one from the output shaft to the cooling fan. Aside from transmitting power and torque, the coupling compensates for misalignments between the two shafts. Misalignment may occur from initial assembly inaccuracies, thermal expansion, shaft deflection or wear [2].
When choosing an appropriate coupling type, it is important to consider the expected misalignment. Rigid couplings are only for a misalignment that occurs during installation and is persistent during operation. On the other hand, flexible couplings can accommodate misalignments that occur during operation [3]. For the application at hand, rigid couplings are preferred.
Of the rigid coupling types, a flanged coupling is optimal because of its simple structure, heavy-duty use and stable load applications, as is the case here. During the selection of a coupling, it is vital to check that the torque and speed capabilities shown in table I are able to withstand the working conditions.

Fig. 2: Elastic Couplings RNG, Cast Iron from [7]

Based on the aforementioned requirements, two types of elastic couplings have been considered, Radial No Gap (RNG) and Double Expansion Anchor (DXA). The difference between these two flanged couplings is the method with which they are mounted onto the shaft. To ease the maintenance of the gear transmission, the RNG flanged coupling has been selected because it is easier to mount and has a better temperature range: -40ºC to +90ºC to prevent thermal expansion [7]. Consider Fig. 2 showing the selected coupling.
Based on the requirements of the gear box shown in table I, two coupling sizes have been selected. For the coupling mounting onto the input shaft, a size 55 and from the output shaft to the cooling fan a size 42 coupling have been selected, based on the following.

$$\begin{aligned}
\text{Input shaft torque} &\rightarrow \overbrace{382[\text{Mm}]}^{\text{Required}} < \overbrace{410\ [\text{Mm}]}^{\text{Rated}}\\
\text{Output shaft torque} &\rightarrow 191[\text{Mm}] < 265\ [\text{Mm}]\\
\text{Input shaft speed} &\rightarrow 1000[\text{rpm}] < 4750\ [\text{rpm}]\\
\text{Output shaft torque} &\rightarrow 2000[\text{rpm}] < 6000\ [\text{rpm}]
\end{aligned}$$

The detailed specifications of these couplings can be seen in Table VII.

C. Shaft

Shafts rotate to transmit power and thereby torque along their length [3]. Unlike standard elements such as gears, shafts are designed from the ground up for specific applications. The way in which a shaft is designed is largely dependant on the mounted elements, the operating loads and service conditions [3].
For the gearbox transmission at hand, the process of designing a shaft is showcased for the output shaft. As shown in Fig. 2 from [1], the shaft is mounted to a fixed structure by two bearings at distance $x = 0$ and $x = L_1$. The pinion gear designed in the previous section is attached to the shaft at $x = L_1/2$ and the coupling to the cooling fan is located at the end of the shaft $x = L_1 + L_2$. For a transmission shaft like this one, a stepped shaft is preferred, that can be designed by looking at the applied loads.

1) Force Analysis: When looking at the mechanical behaviour of the shaft it can be seen that the shaft is twisted, such that the cross-sectional area does not deform, but only rotates [8]. Furthermore, the shaft is considered to be a slender member, meaning the width of the shaft is insignificant in comparison to its length. This allows for the method of sections to be used when solving for the shaft's shear force V and bending moment M.
The loads on the shaft are caused by the mounted elements, and are considered as point forces acting at the midpoint of the element width [3]. There are three forces that the shaft can experience, tangential force F_t, radial force F_r and axial force F_a. These forces are not in the same plane, and all forces must be decomposed into two orthogonal planes, before determining the resultant moment. It is important to note that these loads may be static or fluctuating during operation [3]. The element's weight however can be considered as insignificant and neglected.
When the pinion is rotated by the driving gear, a normal force F_n is exerted at the pitch point of the pinion's teeth. Neglecting the friction force, there are two components of this force, a tangential component and a radial component. Axial force is zero. Consider Fig. 3 showing the force analysis of the output shaft in 3D.

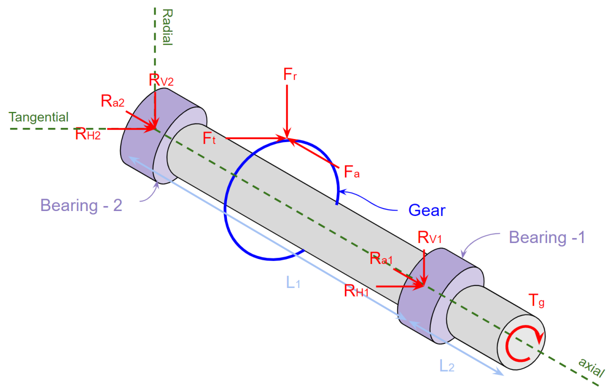

Fig. 3: Force analysis of output shaft in 3D

The magnitude of forces on the pinion and the driving gear are the same but in opposite directions [3]. The assumption that each tooth experiences the repeated loading is also made. Consider the one dimensional set-up of the output shaft shown in Fig. 4, where the two supports represent the bearings.

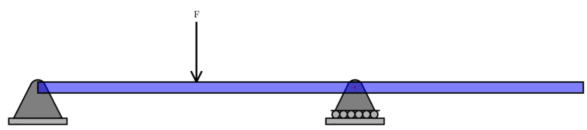

Fig. 4: Set-up of shaft in one plane, made using [9]

Because only one gear is attached to the shaft, in theory the force does not have to be computed in two planes, a plane in the direction of the gear force would also work. But both planes are computed for thoroughness.
In the horizontal direction, there is a tangential force F_t exerted on the shaft by the gear and there are two reaction forces R_H for the two bearings. The tangential force can be calculated based on the given torque and the diameter of the gear using equation 4.

$$\begin{aligned}
F_t &= \frac{2 \cdot 382000}{174} = \frac{2 \cdot 191000}{87}\\
&= 4340\ [\text{N}]
\end{aligned}$$

Now the free body diagram of the set-up can be constructed as shown in Fig. 6. From the free body diagram, the equations for the shear stress $V_t(x)$ are determined as follows:

Range [mm]	Equations of Shear Force :
0 to 125	2170.455
125 to 250	-2170.455
250 to 430	0

Once the shear stress has been obtained, the bending moment $M_t(x)$ can be calculated through the following relation shown in equation 10.

$$V(x) = \frac{dM(x)}{dx} \tag{10}$$

Range [mm]	Equations of Bending Moment:
0 to 125	$2170.455x$
125 to 250	$-2170.455x + 542431.318$
250 to 430	0

The V-line and M-line are plotted and shown in Fig.5

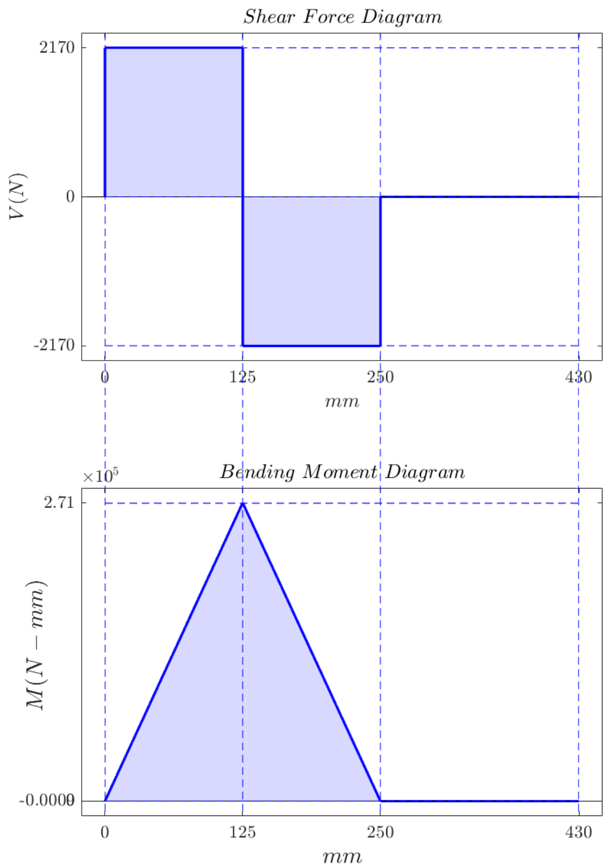

Fig. 5: Shear force and bending moment diagram for horizontal plane, made using [9]

The same procedure as for the horizontal direction is conducted for the vertical direction, with the radial force F_r. The radial force is determined using the known pressure angle $\alpha = 20°$shown in equation 11.

$$F_r = F_t \tan(\alpha) \tag{11}$$

$$\begin{aligned} F_r &= 4340 \cdot \tan(20 \cdot \left(\frac{\pi}{180}\right)) \\ &= 1580 \text{ [N]} \end{aligned}$$

Using the determined radial force, a free body diagram is plotted, as shown in Fig. 7. From this diagram, the following equations and their respective graphs in Fig. 8

Range [mm]	Equations of Shear Force:
0 to 125	789.981
125 to 250	-789.981
250 to 430	0

Range [mm]	Equations of Bending Moment:
0 to 125	$789.981x$
125 to 250	$-789.981x + 197428.854$
250 to 430	0

Now that the force and moment diagrams have been drawn, the following values can be gathered.

$$\begin{aligned} R_{H1} &= R_{H2} = 2170 \text{ [N]} \\ R_{V1} &= R_{V2} = 790 \text{ [N]} \\ M_{H_{MAX}} &= 2.17 \times 10^5 \text{ [Nmm]} \\ M_{V_{MAX}} &= 9.87 \times 10^4 \text{ [Nmm]} \end{aligned}$$

The goal of producing the bending moment diagrams is to determine the distribution of bending moments in the shaft. Looking at the shear and moment diagrams, the “worst” case scenario for the shaft diameter D is at the position of the spur gear. The moments in both planes are maximal and there is torque. Because two planes are used, the total moment acting on the shaft at the point of the spur gear can be calculated using vector summation, shown below:

$$\begin{aligned} M_{MAX} &= \sqrt{M_{V_{MAX}}^2 + M_{H_{MAX}}^2} \\ &= \sqrt{(2.17 \times 10^5)^2 + (9.87 \times 10^4)^2} \\ &= 2.384 \times 10^5 \text{ [Nmm]} \end{aligned}$$

To conclude, the torque line can be plotted as shown in Fig. 9.

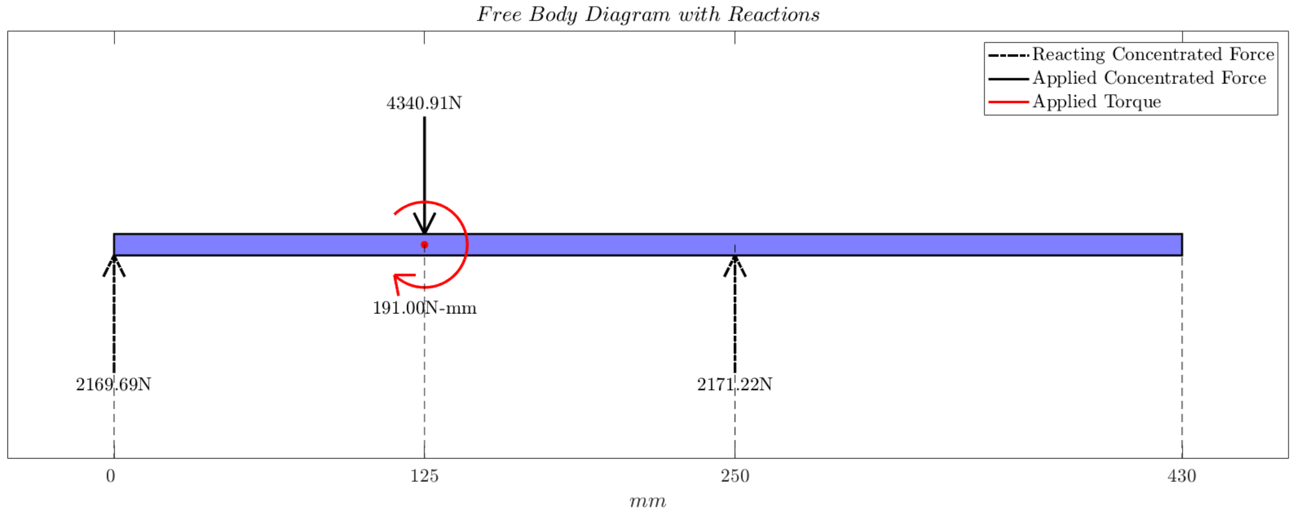

Fig. 6: Free body diagram of horizontal plane, made using [9]

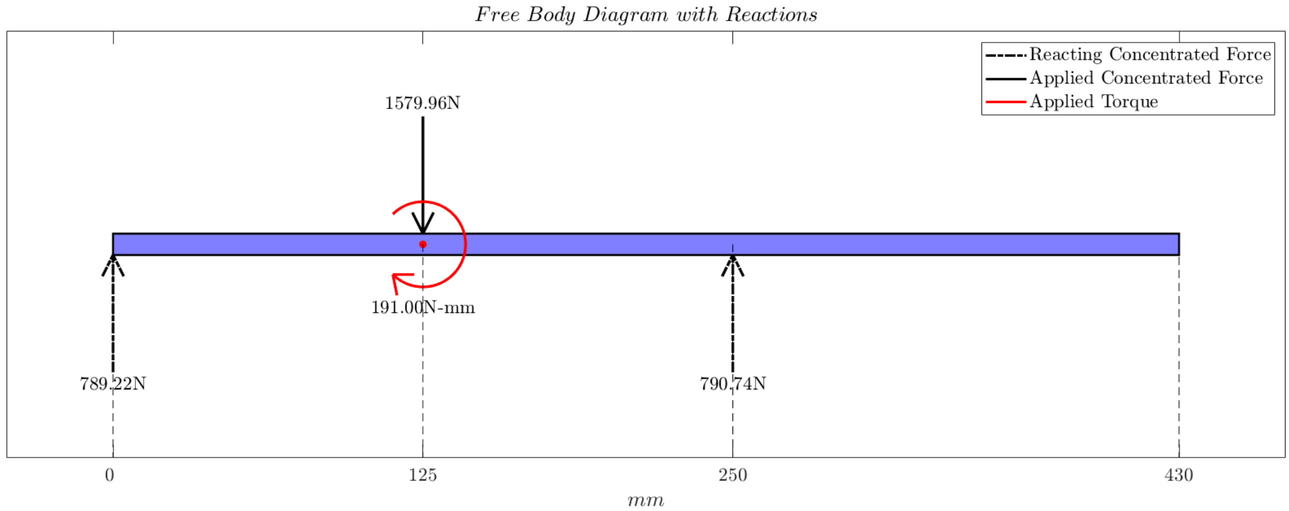

Fig. 7: Free body diagram of vertical plane, made using [9]

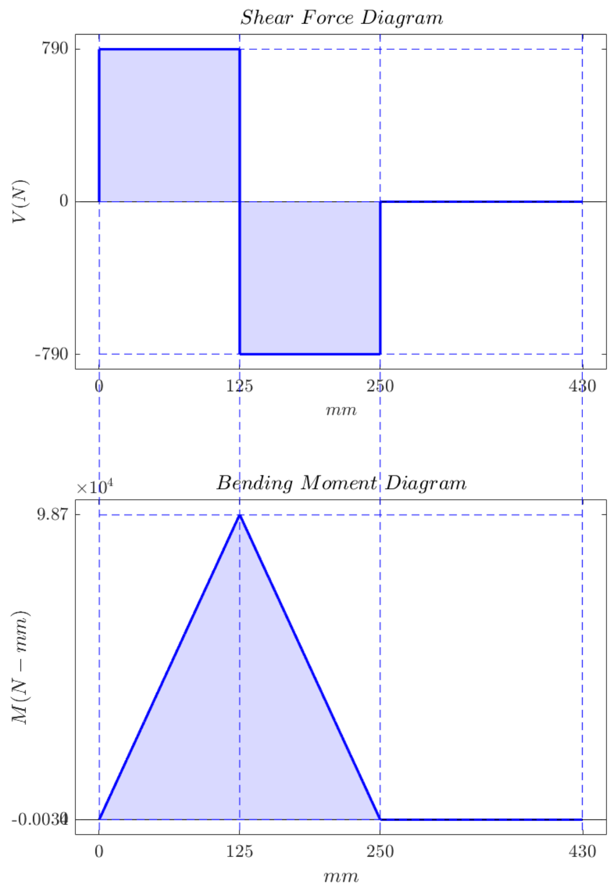

Fig. 8: Shear force and bending moment diagram for vertical plane, made using [9]

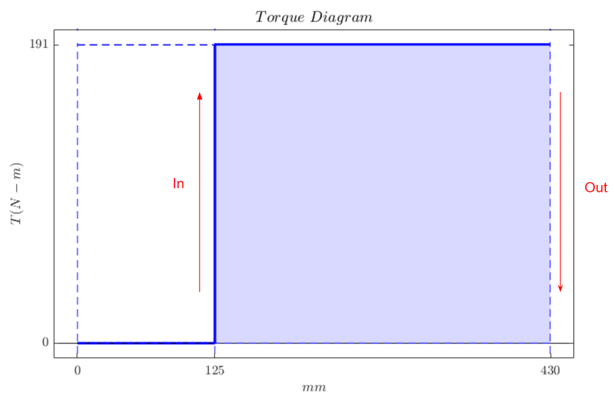

Fig. 9: Torque line for both planes, made using [9]

2) Geometrical Calculations: Consider the proposed geometry of the output shaft shown in Fig. 10. Sharp fillets are located at r_1, r_3, r_5, and well-rounded fillets at, r_2, r_4. The reason for this is explained in the section Mounting Maintenance. Sharp fillets have a stress concentration factor K_t of 2.5 while well-rounded have a 1.5.

The minimum acceptable diameter is then computed based on the diagrams determined in the force analysis. Consider the following sections:
Point A: The first bearing is located at point A, and has a sharp fillet on the right. This is to allow the bearing to slide onto the shaft easily and be pressed to its final position. Normally a light press fit bearing bore, and the shaft seat should be enough [10].
At this point along the shaft length, there is no bending moment and no torque. This means that only the vertical shear stress from the reaction forces is present. The diameter is given by equation 12 [2].

$$D = \sqrt{\frac{2.94K_t \cdot V \cdot K_A}{\tau_{-1}}} \tag{12}$$

To solve this equation the resultant shear stress V is found as follows:

$$\begin{aligned} V &= \sqrt{(790)^2 + (2170)^2} \\ &= 2.301 \times 10^3 \text{ [N]} \end{aligned}$$

The endurance limit in shear τ_{-1} is taken from table 10.4 from [3], for carbon steel Cr45. This material was chosen for reasons mentioned in the section Assumptions and Decisions. Next, a service factor K_A of 1.0 was chosen from table 2.1 from [3] for this application type. Then the diameter was determined as follows.

$$\begin{aligned} D_1 &= \sqrt{\frac{2.94 \cdot 2.5 \cdot 2.301 \times 10^3 \cdot 1.0}{140}} \\ &= 8.5 \text{ [mm]} \end{aligned}$$

This value is rather small and may have to be increased to allow for the bearing to attach.
Point B: is the location of the pinion with a well-rounded fillet to the left, a profile keyseat at the gear, and a sharp fillet to the right. The point of using a well-rounded fillet here is for the bore of the gear to accommodate a large fillet. This means that a chamfer is produced at the ends of the bore [10]. The diameter is found using equation 13.

$$D = \left[\left(\frac{32K_A}{\pi} \right) \sqrt{\left(\frac{K_t M_{MAX}}{\sigma_{-1}} \right)^2 + \frac{3}{4} \left(\frac{T}{\sigma_S} \right)^2} \right]^{1/3} \tag{13}$$

For diameter D_2, the well-rounded fillet stress factor is selected. The yield strength σ_S and endurance limit in normal σ_{-1} taken from table 10.4 from [3], for carbon steel Cr45.

$$\begin{aligned} D_2 = &\left[\left(\frac{32 \cdot 1}{\pi} \right) \sqrt{\left(\frac{1.5 \cdot 2.384 \times 10^5}{172} \right)^2 + \frac{3}{4} \left(\frac{191000}{300} \right)^2} \right]^{1/3} \\ &= 68.8 \text{[mm]} \end{aligned}$$

For diameter D_3, a K_t is chosen for the sharp fillet, as 2.5. On the other hand, the diameter D_4 is decided during the design

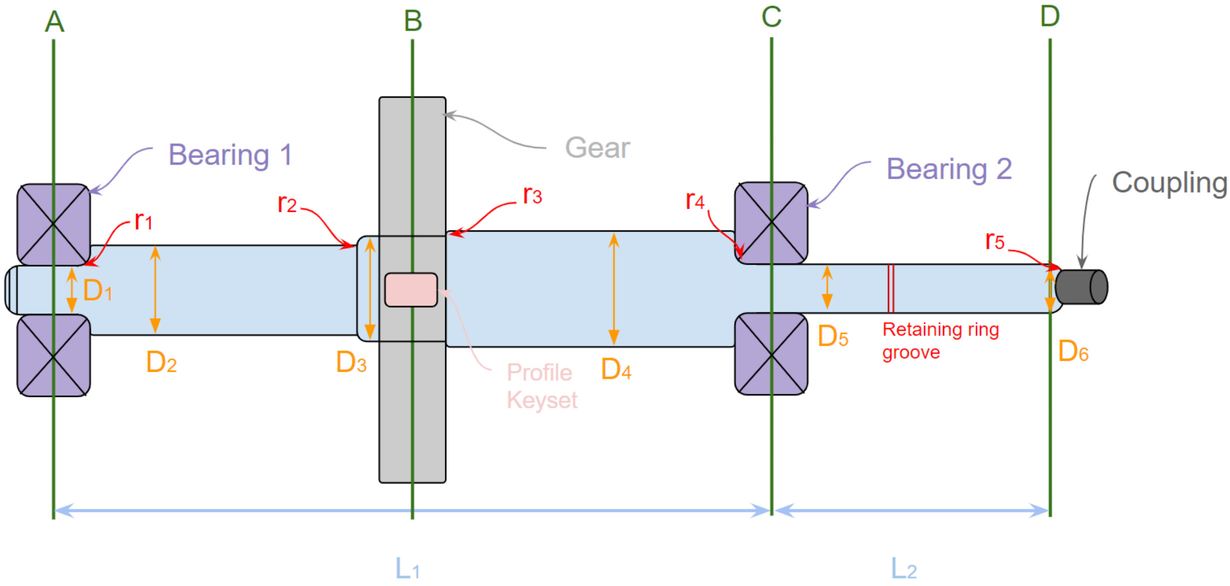

Fig. 10: Proposed geometry for the output shaft. Sharp fillets at r_1, r_3, r_5; well-rounded fillets at , r_2, r_4; profile keyseats at B

of the keys, in order to provide an adequate shoulder for the bearing.

$$D_3 = \left[\left(\frac{32 \cdot 1}{\pi} \right) \sqrt{ \left(\frac{2.5 \cdot 2.384 \times 10^5}{172} \right)^2 + \frac{3}{4} \left(\frac{191000}{300} \right)^2 } \right]^{1/3}$$
$$= 69.5 [\text{mm}]$$

Point C: At this location, there is another bearing, that is supported by a well-rounded fillet and a retaining ring groove on the right-hand side. Due to the ring groove, a K_t value of 3.0 is selected. At this location, the bending moment is zero, but there is a torque present, so equation 13 can be used.

$$D_5 = \left[\left(\frac{32 \cdot 1}{\pi} \right) \sqrt{ \frac{3}{4} \left(\frac{191000}{300} \right)^2 } \right]^{1/3}$$
$$= 17.8 [\text{mm}]$$

Point D: This is where the coupling is attached to the shaft. There is no bending moment or torsion here, so the diameter D_6 is the same as D_1.

Final shaft diameters will be summarized in the section Mounting Maintenance

3) Verification: To verify the strength of the shaft, the combination of bending moment and torque can to considered as a 'worst' case scenario. This combination yields a stress σ_{ca} that is expressed in equation 14.

$$\sigma_{ca} = \frac{\sqrt{M_{MAX}^2 + (\alpha_c T)^2}}{W} \tag{14}$$

The constant α_c is the correction coefficient, that depicts the type of applied load, static or fluctuating. A value of 0.6 has been selected to indicate that it could be fluctuating. In addition, W is the section modulus [mm^3] which, for a symmetrical circular cross-section of the shaft, is found by the second moment of area I.

$$I = \pi \frac{d^4}{64}$$
$$W = 2\frac{I}{d} = \pi \frac{d^3}{32} = \pi \frac{(69.5)^3}{32}$$
$$= 3.230 \times 10^4 \ [\text{mm}^3]$$

Therefore, the combination yields a stress σ_{ca} is lower than endurance limit σ_{-1}, so the shaft is safe.

$$\sigma_{ca} = \frac{\sqrt{2.384 \times 10^5 + (0.6 \cdot 191000)^2}}{3.230 \times 10^4}$$
$$= 8.023 \ [\text{MPa}]$$

D. *Keys*

Keys are elements that enable transmission of torque and power between elements that are attached to a shaft [3]. In this situation, a keyset is needed to mount the gear onto the shaft, and to keep it from moving in the axial direction. A parallel type A keyset has been selected for this application, because it is easy to remove and replace if needed.

Based on minimum diameter D_3 from the shaft calculations, the size of the keyset can be determined. Keys are highly standardized and are selected from existing options rather than designed from scratch [3]. To find the key required, the minimum diameter D_3 is increased by 10 % to account for the removed material [2]. Then, using table 4.1 from [3], the width and height ratio of the keyset is found to be 22×14.

$$D = 1.1 \cdot D_3 = 1.1 \cdot 69.50 = 76.5 \approx 77 \text{ [mm]}$$
$$b_{key} \times h_{key} = 22 \times 14$$

Next the base width of the gear which is 90 [mm], is used to determine the key length using table 4.1 [3], which is selected to be 80 [mm]. With these dimensions, the working length of the key l can be determined for type A.

$$\begin{aligned} l_{key} &= L - b_{key} = 80 - 22 \\ &= 58 \text{ [mm]} \end{aligned}$$

1) Stress Analysis: The failure mode for key connections is the crushing of the weakest material. For a steel gear on a shaft under stable loads, an allowable bearing stress $|\sigma_p|$ of 80 [MPa] is selected.
To ensure safety, the key must be able to withstand the stress given in equation 15.

$$\sigma_p = \frac{4T}{h_{key} l_{key} D_3} \tag{15}$$

$$\begin{aligned} \sigma_p &= \frac{4 \cdot 191000}{14 \cdot 58 \cdot 69.50} \\ &= 13.53 \text{ [MPa]} \end{aligned}$$

The stress is significantly lower than the allowable bearing stress, meaning it is safe. Furthermore, the stress could be doubled, and the key would not suffer a failure.
Another key has been calculated in the same process to attach the coupling. The final dimensions are 6x6x34. These can be seen in Table VII.

E. Bearings

Rolling bearing support rotating shafts while allowing motion between the two elements [3]. The two bearings located on the shaft seen in Fig. 10 are mounted in a fixed-floating orientation. This means that the fixed bearing, on the left carries a radial load and all the axial loads, while the floating bearing carries only radial loads [2]. Even thought the spur gear does not exert an axial load on the shaft and there is no other direct source of axial loads, it is vital to must account for unexpected axial loads such as thermal expansion or manufacturing tolerances, through this orientation [2].
The type of bearing that is considered is a deep groove ball bearing or a cylindrical bearing. This is largely due to the lack of axial forces. However, the required load carrying capacity will be the deciding factor between the two types of bearings.

1) Life expectancy and load carrying capacities: Bearings are highly standardized, and are selected based on their load capacities and intended lifetime. Considering the element's lifetime is more important for bearings than in other elements such as spur gears or shaft's because of the large amount of failure modes that can occur over time. For instance, fretting, fatigue and corrosion [2].
The expected lifetime of 5 years is used to determine the dynamic load capacity C using equation 16.

$$C = \frac{f_p P}{f_t} \left(\frac{60 n L_h}{10^6 \alpha_a} \right)^{1/\epsilon} \tag{16}$$

In this equation, f_p and f_t are friction forces, which have previously been assumed to be zero. Furthermore, the lifetime in hours L_h has already been calculated to be 24000 hours, and the speed n is given. The goal is to ensure safety, which is why a 99 % reliability factor has been chosen. It is expressed in terms of $\alpha_a = 0.21$ [3]. Lastly, P is the equivalent dynamic load. It is determined by the following equation 17, where R_L is the radial force, A_L the axial force and X, Y are their coefficients.

$$P = X R_L + Y A_L \tag{17}$$

As mentioned already, the system at hand experiences no direct axial loads and a radial load distribution is assumed. Therefore, A_L is zero for both bearings. Because of this, the ratio A/C_0 referred to the radial axial load is less than factor e. For this reason, the radial load factor X is equal to one. Now solving equation 16 for ball bearings case $\epsilon = 3$.

$$\begin{aligned} P &= V = 2.301 \times 10^3 \text{ [N]} \\ C &= 2.301 \times 10^3 \left(\frac{60 \cdot 1000 \cdot 24000}{10^6 \cdot 0.21} \right)^{1/3} \\ &= 4.387 \times 10^4 \text{ [N]} = 43.87 \text{ [kN]} \end{aligned}$$

Due to the high dynamic load rating, a cylindrical rolling bearing has been selected from [7]. The chosen bearing is a NU 206 ECP with $C = 44.0$ [kN], and it can be seen in Fig. 11 below:

Fig. 11: Cylindrical Roller Bearings SKF NU 206 ECP from [7]

2) verification: To verify that the bearing is sufficient, the dynamic load rating is used to determine the lifetime by rewriting equation 16.

$$L_h = \frac{10^6 \cdot 0.21}{60 \cdot 1000} \left(\frac{44000}{2.301 \times 10^3} \right)^3$$
$$= 24472 \text{ [h]}$$

Therefore, the bearing will survive at least an additional 472 hours. Another verification that needs to be made, is to check that the diameter is larger than the minimum diameter of the shaft. The diameter is 30mm meaning it is larger than both D_1 and D_5.

F. *Mounting & Maintenance*

Now that all the components of the output shaft have been analysed and designed, the following design decisions are made. On the right-hand side of the gear, and the first bearing, a shoulder is intentionally manufactured. This is to allow for the shaft and key to attach the gear compactly. At these sharp fillets, the recommend ISO fits of R1.4 are placed, while the well-rounded fillets are R2 [6].
To prevent further stress concentrations, the keyset is located at the centre of the gear instead of near the side with the shoulder. Additionally, corner chamfers are added to help with manufacturing [2].
For the sections of the shaft where the gear and coupling are placed, a high tolerance is required, unlike in the bearings. This is because the bearings are in a fixed-floating set-up.
The final shaft diameters are depicted in table VI. All the actual diameters are not less than the minimum while being as low as they can. The diameter at D_1, D_5 and D_7 is set by the bearings and coupling. Additionally, D_4 has an additional 5mm added to it to create a shoulder. Besides that, the minimum diameter is used.

Diameter	Maitng part	Minimum diameter [mm]	Actual diameter [mm]
D_1	Bearing	8.5	30.0
D_2	Nothing	68.8	68.8
D_3	Gear	69.5	69.5
D_4	Nothing	68.8	74.5
D_5	Bearing	17.8	30.0
D_6	Nothing	17.8	17.8
D_7	Coupling	8.5	13.0

TABLE VI: Shaft diameters [7]

Lastly, the system can be maintained by swapping out the parts. The keysets allow for easy disassembly, and the form fit bearings are fitted onto the shaft from either side allowing for lubrication and disassembly.

V. Technical Drawings and Documentation

Below, two technical drawings can be found. A detailed technical drawing of the output shaft is shown first. It illustrates all the design choices and calculations made in the report so that it can be manufactured. Along with the detailed drawing is an assembly overview of the output shaft with all the components attached to it. These drawings conclude the design process.

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VII. Appendix

A. Dimensions

Coupling					
Size	**Torque Nominal[Nm]**	**Torque max. [Nm]**	**d Pilot Bore [mm]**	**d max. [mm]**	**Max. Speed at 30 m/s [rpm]**
55	410	820	16	55	4750
75	1280	2560	25	80	3550
Bearing					
d [mm]	**D [mm]**	**B [mm]**	**Displacement [Direction]**	**Load Rating dyn. C [kN]**	**Load Rating stat. C0 [kN]**
30	62	16	one-sided	44,0	36,5
Keys					
width [mm]		**height [mm]**		**length [mm]**	
22		14		58	
6		6		34	

TABLE VII: Dimensions of chosen parts from [7]

B. Load Factor calculation for gears

The modified pitch diameter by actual load factor K_a is given by equation 18.

$$d_{pa} = d_p \sqrt[3]{K_{Ha}/K} \tag{18}$$

Here the load factor K_{Ha} is given by equation 20.

$$K = K_A K_V K_{H\alpha} K_{H\beta} \tag{19}$$

For a pitch line velocity $v = 4.51$ [m/s], and material gear accuracy grade 7, K_v is given by:

$$K_v = \left(1 + \frac{\sqrt{200v}}{A}\right)^B$$

$$B = 0.25(Q-5)^{0.667} = 0.25(7-5)^{0.667} = 0.397$$

$$A = 50 + 56(1.0 - B) = 50 + 56(1.0 - 0.397) = 83.768$$

$$K_v = \left(1 + \frac{\sqrt{200 \cdot 4.51}}{83.768}\right)^0 .397$$

$$= 1.127$$

From table 8.2 from [3], $K_{H\alpha} = 1$, and from table 2.1 [3], $K_A = 1$. Next, $K_{H\beta}$ is found from table 8.3 as follows:

$$\begin{aligned} K_{H\beta} &= a_1 + a_2\left(1 + a_3\varphi_d^2\right)\varphi_d^2 + a_4 b \\ &= 1.12 + 0.18(1) \cdot 1^2 + 2.3 \cdot 10^{-4} \cdot 94.82 \\ &= 1.359 \end{aligned}$$

Therefore:

$$K_{Ha} = 1 \cdot 1.127 \cdot 1 \cdot 1.359$$
$$= 1.532$$

Next the load factor K_{Fa} is given by equation 20.

$$K = K_A K_V K_{F\alpha} K_{F\beta} \tag{20}$$

From table 8.2 from [3], $K_{F\alpha} = 1$, and $K_{F\beta}$ is found from table 8.3 where a standard tooth height of 2.25mm is chosen

$$N_0 = \frac{(b/h)^2}{1 + (b/h) + (b/h)^2}$$
$$= \frac{14.178^2}{1 + 14.178 + 14.178^2}$$
$$= 0.930$$

$$K_{F\beta} = (K_{H\beta})^{N_0} = 1.314^{0.930} = 1.330$$

Therefore:

$$K_{Fa} = 1 \cdot 1.127 \cdot 1 \cdot 1.330$$
$$= 1.499$$

Lastly:

$$d_{pa} = 86.20 \sqrt[3]{1.532/1.3}$$
$$= 87.75 \text{ [mm]}$$